

Question ID 1efd7ef3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 1efd7ef3

What is the value of $\sin 42\pi$?

- A. 0
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. 1

ID: 1efd7ef3 Answer

Correct Answer:

A

Rationale

Choice A is correct. The sine of a number t is the y -coordinate of the point arrived at by traveling a distance of t units counterclockwise around the unit circle from the starting point $(1, 0)$. Since the unit circle has a circumference of 2π units, it follows that one full rotation around the circle is equal to a distance of 2π units. Therefore, a distance of 42π units around the circle from the starting point $(1, 0)$ would result in exactly **21** full rotations, arriving back at the point $(1, 0)$. So, $\sin 42\pi$ is equal to the y -coordinate of the point $(1, 0)$, which is **0**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $\cos 42\pi$, not $\sin 42\pi$.

Question Difficulty:

Hard

Question ID 6585d841

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 6585d841

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's radius?

- A. $\sqrt{109}$
- B. $\sqrt{149}$
- C. $\sqrt{167}$
- D. $\sqrt{341}$

ID: 6585d841 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that in the xy -plane, the graph of the given equation is a circle. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the length of the circle's radius. Subtracting $6y$ from both sides of the equation $x^2 + 14x + y^2 = 6y + 109$ yields $x^2 + 14x + y^2 - 6y = 109$. By completing the square, this equation can be rewritten as $\left(x^2 + 14x + \left(\frac{14}{2}\right)^2\right) + \left(y^2 - 6y + \left(\frac{-6}{2}\right)^2\right) = 109 + \left(\frac{14}{2}\right)^2 + \left(\frac{-6}{2}\right)^2$. This equation can be rewritten as $(x^2 + 14x + 49) + (y^2 - 6y + 9) = 109 + 49 + 9$, or $(x + 7)^2 + (y - 3)^2 = 167$. Therefore, $r^2 = 167$. Taking the square root of both sides of this equation yields $r = \sqrt{167}$ and $r = -\sqrt{167}$. Since r is the length of the circle's radius, r must be positive. Therefore, the length of the circle's radius is $\sqrt{167}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID acd30391

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: acd30391

A circle in the xy -plane has equation $(x + 3)^2 + (y - 1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

- A. $(-7, 3)$
- B. $(-3, 1)$
- C. $(0, 0)$
- D. $(3, 2)$

ID: acd30391 Answer

Correct Answer:
D

Rationale

Choice D is correct. The circle with equation $(x + 3)^2 + (y - 1)^2 = 25$ has center $(-3, 1)$ and radius 5. For a point to be inside of the circle, the distance from that point to the center must be less than the radius, 5. The distance between $(3, 2)$ and $(-3, 1)$ is $\sqrt{(-3 - 3)^2 + (1 - 2)^2} = \sqrt{(-6)^2 + (-1)^2} = \sqrt{37}$, which is greater than 5. Therefore, $(3, 2)$ does NOT lie in the interior of the circle.

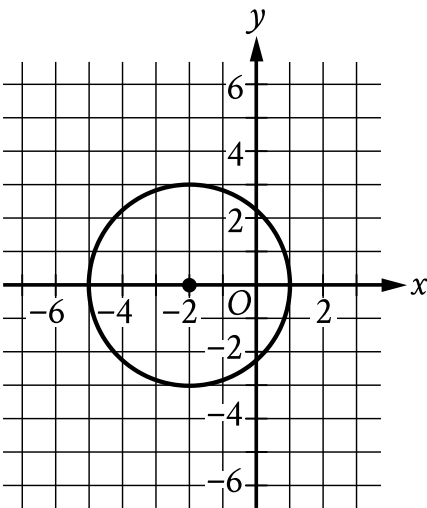
Choice A is incorrect. The distance between $(-7, 3)$ and $(-3, 1)$ is $\sqrt{(-7 + 3)^2 + (3 - 1)^2} = \sqrt{(-4)^2 + (2)^2} = \sqrt{20}$, which is less than 5, and therefore $(-7, 3)$ lies in the interior of the circle. Choice B is incorrect because it is the center of the circle. Choice C is incorrect because the distance between $(0, 0)$ and $(-3, 1)$ is $\sqrt{(0 + 3)^2 + (0 - 1)^2} = \sqrt{(3)^2 + (1)^2} = \sqrt{8}$, which is less than 5, and therefore $(0, 0)$ is in the interior of the circle.

Question Difficulty:
Hard

Question ID a38c0183

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: a38c0183



Circle A (shown) is defined by the equation $(x + 2)^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. Which equation defines circle B?

- A. $(x + 2)^2 + (y + 6)^2 = (4)(9)$
- B. $2(x + 2)^2 + 2(y + 6)^2 = 9$
- C. $(x + 2)^2 + (y - 6)^2 = (4)(9)$
- D. $2(x + 2)^2 + 2(y - 6)^2 = 9$

ID: a38c0183 Answer

Correct Answer:

A

Rationale

Choice A is correct. According to the graph, the center of circle A has coordinates $(-2, 0)$, and the radius of circle A is 3. It's given that circle B is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. It follows that the center of circle B is 6 units below the center of circle A. The point that's 6 units below $(-2, 0)$ has the same x-coordinate as $(-2, 0)$ and has a y-coordinate that is 6 less than the y-coordinate of $(-2, 0)$. Therefore, the coordinates of the center of circle B are $(-2, 0 - 6)$, or $(-2, -6)$. Since the radius of circle B is 2 times the radius of circle A, the radius of circle B is $(2)(3)$. A circle in the xy-plane can be defined by an equation of the form $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the radius of the circle is r . Substituting -2 for h , -6 for k , and $(2)(3)$ for r in this equation yields $(x - (-2))^2 + (y - (-6))^2 = ((2)(3))^2$, which is equivalent to $(x + 2)^2 + (y + 6)^2 = (2)^2(3)^2$, or $(x + 2)^2 + (y + 6)^2 = (4)(9)$. Therefore, the equation $(x + 2)^2 + (y + 6)^2 = (4)(9)$ defines circle B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation defines a circle that's the result of shifting circle A up, not down, by **6** units and increasing the radius.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID fb58c0db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: fb58c0db

Points A and B lie on a circle with radius 1, and arc $\overline{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overline{AB}$?

ID: fb58c0db Answer

Rationale

The correct answer is $\frac{1}{6}$. The circumference, C , of a circle is $C = 2\pi r$, where r is the length of the radius of the circle. For the given circle with a radius of 1, the circumference is $C = 2(\pi)(1)$, or $C = 2\pi$. To find what fraction of the circumference the length of arc $\overline{AB}$ is, divide the length of the arc by the circumference, which gives $\frac{\pi}{3} \div 2\pi$. This division can be represented by $\frac{\pi}{3} \cdot \frac{1}{2\pi} = \frac{1}{6}$. Note that 1/6, .1666, .1667, 0.166, and 0.167 are examples of ways to enter a correct answer.

Question Difficulty:
Hard

Question ID e80d62c6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: e80d62c6

The equation $x^2 + (y - 2)^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the xy -plane. Which of the following equations represents circle B?

- A. $x^2 + (y + 2)^2 = 36$
- B. $x^2 + (y - 6)^2 = 36$
- C. $(x - 4)^2 + (y - 2)^2 = 36$
- D. $(x + 4)^2 + (y - 2)^2 = 36$

ID: e80d62c6 Answer

Correct Answer:
A

Rationale

Choice A is correct. The standard form of an equation of a circle in the xy -plane is $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the length of the radius of the circle is r . The equation of circle A, $x^2 + (y - 2)^2 = 36$, can be rewritten as $(x - 0)^2 + (y - 2)^2 = 6^2$. Therefore, the center of circle A is at $(0, 2)$ and the length of the radius of circle A is 6. If circle A is shifted down 4 units, the y -coordinate of its center will decrease by 4; the radius of the circle and the x -coordinate of its center will not change. Therefore, the center of circle B is at $(0, 2 - 4)$, or $(0, -2)$, and its radius is 6. Substituting 0 for h , -2 for k , and 6 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 0)^2 + (y - (-2))^2 = (6)^2$, or $x^2 + (y + 2)^2 = 36$. Therefore, the equation $x^2 + (y + 2)^2 = 36$ represents circle B.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A up, rather than down, 4 units.

Choice C is incorrect. This equation represents a circle obtained by shifting circle A right, rather than down, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A left, rather than down, 4 units.

Question Difficulty:
Hard

Question ID 196e8e6e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 196e8e6e

In the xy -plane, a circle has center C with coordinates (h, k) . Points A and B lie on the circle. Point A has coordinates $(h + 1, k + \sqrt{102})$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

- A. $\sqrt{206}$
- B. $2\sqrt{102}$
- C. $103\sqrt{2}$
- D. $103\sqrt{3}$

ID: 196e8e6e Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that points A and B lie on the circle with center C . Therefore, $\overline{AC}$ and $\overline{BC}$ are both radii of the circle. Since all radii of a circle are congruent, $\overline{AC}$ is congruent to $\overline{BC}$. The length of $\overline{AC}$, or the distance from point A to point C , can be found using the distance formula, which gives the distance between two points, (x_1, y_1) and (x_2, y_2) , as $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. Substituting the given coordinates of point A , $(h + 1, k + \sqrt{102})$, for (x_1, y_1) and the given coordinates of point C , (h, k) , for (x_2, y_2) in the distance formula yields $\sqrt{(h + 1 - h)^2 + (k + \sqrt{102} - k)^2}$, or $\sqrt{1^2 + (\sqrt{102})^2}$, which is equivalent to $\sqrt{1 + 102}$, or $\sqrt{103}$. Therefore, the length of $\overline{AC}$ is $\sqrt{103}$ and the length of $\overline{BC}$ is $\sqrt{103}$. It's given that angle ACB is a right angle. Therefore, triangle ACB is a right triangle with legs $\overline{AC}$ and $\overline{BC}$ and hypotenuse $\overline{AB}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting $\sqrt{103}$ for a and b in this equation yields $(\sqrt{103})^2 + (\sqrt{103})^2 = c^2$, or $103 + 103 = c^2$, which is equivalent to $206 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{206} = c$. Therefore, the length of $\overline{AB}$ is $\sqrt{206}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the length of $\overline{AB}$ if the length of $\overline{AC}$ were 103 , not $\sqrt{103}$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID 89661424

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 89661424

A circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9 . An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

ID: 89661424 Answer

Correct Answer:
-52

Rationale

The correct answer is -52 . The equation of a circle in the xy -plane with its center at (h, k) and a radius of r can be written in the form $(x - h)^2 + (y - k)^2 = r^2$. It's given that a circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9 . Substituting -5 for h , 2 for k , and 9 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - (-5))^2 + (y - 2)^2 = 9^2$, or $(x + 5)^2 + (y - 2)^2 = 81$. It's also given that an equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. Therefore, $(x + 5)^2 + (y - 2)^2 = 81$ can be rewritten in the form $x^2 + y^2 + ax + by + c = 0$. The equation $(x + 5)^2 + (y - 2)^2 = 81$, or $(x + 5)(x + 5) + (y - 2)(y - 2) = 81$, can be rewritten as $x^2 + 5x + 5x + 25 + y^2 - 2y - 2y + 4 = 81$. Combining like terms on the left-hand side of this equation yields $x^2 + y^2 + 10x - 4y + 29 = 81$. Subtracting 81 from both sides of this equation yields $x^2 + y^2 + 10x - 4y - 52 = 0$, which is equivalent to $x^2 + y^2 + 10x + (-4)y + (-52) = 0$. This equation is in the form $x^2 + y^2 + ax + by + c = 0$. Therefore, the value of c is -52 .

Question Difficulty:
Hard

Question ID 981275d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 981275d2

$(x - 6)^2 + (y + 5)^2 = 16$

In the xy -plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

- A. $(2, -5)$
- B. $(6, -1)$
- C. $(6, -5)$
- D. $(6, -9)$

ID: 981275d2 Answer

Correct Answer:
A

Rationale

Choice A is correct. The standard form for the equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) are the coordinates of the center and r is the length of the radius. According to the given equation, the center of the circle is $(6, -5)$. Let (x_1, y_1) represent the coordinates of point Q . Since point $P(10, -5)$ and point $Q(x_1, y_1)$ are the endpoints of a diameter of the circle, the center $(6, -5)$ lies on the diameter, halfway between P and Q . Therefore, the following relationships hold: $\frac{x_1 + 10}{2} = 6$ and $\frac{y_1 + (-5)}{2} = -5$. Solving the equations for x_1 and y_1 , respectively, yields $x_1 = 2$ and $y_1 = -5$. Therefore, the coordinates of point Q are $(2, -5)$.

Alternate approach: Since point $P(10, -5)$ on the circle and the center of the circle $(6, -5)$ have the same y -coordinate, it follows that the radius of the circle is $10 - 6 = 4$. In addition, the opposite end of the diameter $\overline{PQ}$ must have the same y -coordinate as P and be 4 units away from the center. Hence, the coordinates of point Q must be $(2, -5)$.

Choices B and D are incorrect because the points given in these choices lie on a diameter that is perpendicular to the diameter $\overline{PQ}$. If either of these points were point Q , then $\overline{PQ}$ would not be the diameter of the circle. Choice C is incorrect because $(6, -5)$ is the center of the circle and does not lie on the circle.

Question Difficulty:
Hard

Question ID 9d159400

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 9d159400

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

- A. $(x - 8)^2 + (y - 8)^2 = 16$
- B. $(x - 8)^2 + (y - 4)^2 = 16$
- C. $(x - 4)^2 + (y - 9)^2 = 16$
- D. $x^2 + (y - 9)^2 = 16$

ID: 9d159400 Answer

Correct Answer:

C

Rationale

Choice C is correct. The graph of the equation $(x - h)^2 + (y - k)^2 = r^2$ in the xy -plane is a circle with center (h, k) and a radius of length r . The radius of a circle is the distance from the center of the circle to any point on the circle. If a circle in the xy -plane intersects the y -axis at exactly one point, then the perpendicular distance from the center of the circle to this point on the y -axis must be equal to the length of the circle's radius. It follows that the x -coordinate of the circle's center must be equivalent to the length of the circle's radius. In other words, if the graph of $(x - h)^2 + (y - k)^2 = r^2$ is a circle that intersects the y -axis at exactly one point, then $r = |h|$ must be true. The equation in choice C is $(x - 4)^2 + (y - 9)^2 = 16$, or $(x - 4)^2 + (y - 9)^2 = 4^2$. This equation is in the form $(x - h)^2 + (y - k)^2 = r^2$, where $h = 4$, $k = 9$, and $r = 4$, and represents a circle in the xy -plane with center $(4, 9)$ and radius of length 4. Substituting 4 for r and 4 for h in the equation $r = |h|$ yields $4 = |4|$, or $4 = 4$, which is true. Therefore, the equation in choice C represents a circle in the xy -plane that intersects the y -axis at exactly one point.

Choice A is incorrect. This is the equation of a circle that does not intersect the y -axis at any point.

Choice B is incorrect. This is an equation of a circle that intersects the x -axis, not the y -axis, at exactly one point.

Choice D is incorrect. This is the equation of a circle with the center located on the y -axis and thus intersects the y -axis at exactly two points, not exactly one point.

Question Difficulty:

Hard

Question ID ca2235f6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: ca2235f6

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

- A. 3
- B. 6
- C. 9
- D. 24

ID: ca2235f6 Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that the measure of arc AB is 45° and the length of arc AB is 3 inches. The arc measure of the full circle is 360° . If x represents the circumference, in inches, of the circle, it follows that $\frac{45^\circ}{360^\circ} = \frac{3 \text{ inches}}{x \text{ inches}}$. This equation is equivalent to $\frac{45}{360} = \frac{3}{x}$, or $\frac{1}{8} = \frac{3}{x}$. Multiplying both sides of this equation by $8x$ yields $1(x) = 3(8)$, or $x = 24$. Therefore, the circumference of the circle is 24 inches.

Choice A is incorrect. This is the length of arc AB .

Choice B is incorrect and may result from multiplying the length of arc AB by 2 .

Choice C is incorrect and may result from squaring the length of arc AB .

Question Difficulty:
Hard

Question ID 0acfddeb5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 0acfddeb5

A circle has center G , and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N , respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral $GMHN$ is 3,856 millimeters, what is the distance, in millimeters, between points G and H ?

- A. 168
- B. 1,752
- C. 1,760
- D. 1,768

ID: 0acfddeb5 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the radius of the circle is 168 millimeters. Since points M and N both lie on the circle, segments GM and GN are both radii. Therefore, segments GM and GN each have length 168 millimeters. Two segments that are tangent to a circle and have a common exterior endpoint have equal length. Therefore, segment MH and segment NH have equal length. Let x represent the length of segment MH . Then x also represents the length of segment NH . It's given that the perimeter of quadrilateral $GMHN$ is 3,856 millimeters. Since the perimeter of a quadrilateral is equal to the sum of the lengths of the sides of the quadrilateral, $3,856 = 168 + 168 + x + x$, or $3,856 = 336 + 2x$. Subtracting 336 from both sides of this equation yields $3,520 = 2x$, and dividing both sides of this equation by 2 yields $1,760 = x$. Therefore, the length of segment MH is 1,760 millimeters. A line segment that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. Therefore, segment GM is perpendicular to segment MH . Since perpendicular segments form right angles, angle GMH is a right angle. Therefore, triangle GMH is a right triangle with legs of length 1,760 millimeters and 168 millimeters, and hypotenuse GH . By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting 1,760 for a and 168 for b in this equation yields $1,760^2 + 168^2 = c^2$, or $3,125,824 = c^2$. Taking the square root of both sides of this equation yields $\pm 1,768 = c$. Since c represents a length, which must be positive, the value of c is 1,768. Therefore, the length of segment GH is 1,768 millimeters, so the distance between points G and H is 1,768 millimeters.

Choice A is incorrect. This is the distance between points G and M and between points G and N , not the distance between points G and H .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the distance between points M and H and between points N and H , not the distance between points G and H .

Question Difficulty:

Question ID 244ff6c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 244ff6c4

What is the value of $\tan \frac{92\pi}{3}$?

- A. $-\sqrt{3}$
- B. $-\frac{\sqrt{3}}{3}$
- C. $\frac{\sqrt{3}}{3}$
- D. $\sqrt{3}$

ID: 244ff6c4 Answer

Correct Answer:

A

Rationale

Choice A is correct. A trigonometric ratio can be found using the unit circle, that is, a circle with radius **1** unit. If a central angle of a unit circle in the xy -plane centered at the origin has its starting side on the positive x -axis and its terminal side intersects the circle at a point (x, y) , then the value of the tangent of the central angle is equal to the y -coordinate divided by the x -coordinate. There are 2π radians in a circle. Dividing $\frac{92\pi}{3}$ by 2π yields $\frac{92}{6}$, which is equivalent to $15 + \frac{2}{3}$. It follows that on the unit circle centered at the origin in the xy -plane, the angle $\frac{92\pi}{3}$ is the result of **15** revolutions from its starting side on the positive x -axis followed by a rotation through $\frac{2\pi}{3}$ radians. Therefore, the angles $\frac{92\pi}{3}$ and $\frac{2\pi}{3}$ are coterminal angles and $\tan(\frac{92\pi}{3})$ is equal to $\tan(\frac{2\pi}{3})$. Since $\frac{2\pi}{3}$ is greater than $\frac{\pi}{2}$ and less than π , it follows that the terminal side of the angle is in quadrant II and forms an angle of $\frac{\pi}{3}$, or **60°**, with the negative x -axis. Therefore, the terminal side of the angle intersects the unit circle at the point $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$. It follows that the value of $\tan(\frac{2\pi}{3})$ is $\frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}$, which is equivalent to $-\sqrt{3}$. Therefore, the value of $\tan(\frac{92\pi}{3})$ is $-\sqrt{3}$.

Choice B is incorrect. This is the value of $\frac{1}{\tan(\frac{92\pi}{3})}$, not $\tan(\frac{92\pi}{3})$.

Choice C is incorrect. This is the value of $\frac{1}{\tan(\frac{\pi}{3})}$, not $\tan(\frac{92\pi}{3})$.

Choice D is incorrect. This is the value of $\tan(\frac{\pi}{3})$, not $\tan(\frac{92\pi}{3})$.

Question Difficulty:

Hard

Question ID 9acd101f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 9acd101f

The equation $x^2 + (y - 1)^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down 2 units in the xy -plane. Which of the following equations represents circle B?

- A. $(x - 2)^2 + (y - 1)^2 = 49$
- B. $x^2 + (y - 3)^2 = 49$
- C. $(x + 2)^2 + (y - 1)^2 = 49$
- D. $x^2 + (y + 1)^2 = 49$

ID: 9acd101f Answer

Correct Answer:
D

Rationale

Choice D is correct. The graph in the xy -plane of an equation of the form $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center (h, k) and a radius of length r . It's given that circle A is represented by $x^2 + (y - 1)^2 = 49$, which can be rewritten as $x^2 + (y - 1)^2 = 7^2$. Therefore, circle A has center $(0, 1)$ and a radius of length 7. Shifting circle A down two units is a rigid vertical translation of circle A that does not change its size or shape. Since circle B is obtained by shifting circle A down two units, it follows that circle B has the same radius as circle A, and for each point (x, y) on circle A, the point $(x, y - 2)$ lies on circle B. Moreover, if (h, k) is the center of circle A, then $(h, k - 2)$ is the center of circle B. Therefore, circle B has a radius of 7 and the center of circle B is $(0, 1 - 2)$, or $(0, -1)$. Thus, circle B can be represented by the equation $x^2 + (y + 1)^2 = 7^2$, or $x^2 + (y + 1)^2 = 49$.

Choice A is incorrect. This is the equation of a circle obtained by shifting circle A right 2 units.

Choice B is incorrect. This is the equation of a circle obtained by shifting circle A up 2 units.

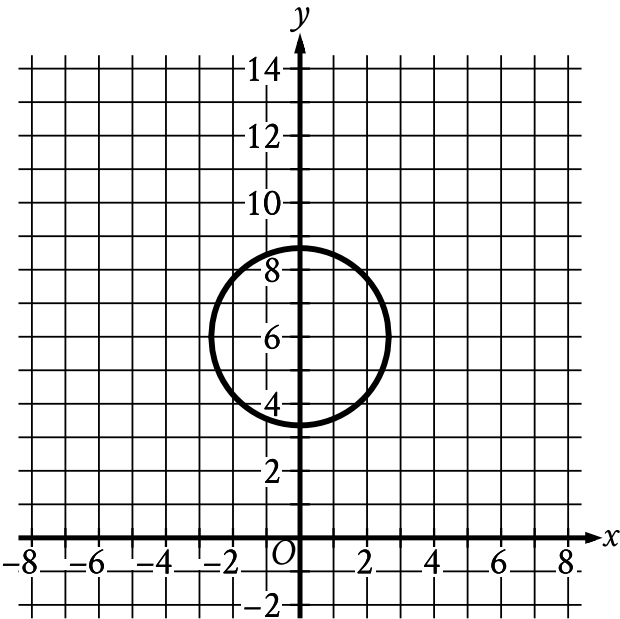
Choice C is incorrect. This is the equation of a circle obtained by shifting circle A left 2 units.

Question Difficulty:
Hard

Question ID 1b2b20b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 1b2b20b9



Circle A shown is defined by the equation $x^2 + (y - 6)^2 = 7$. Circle B (not shown) has the same radius but is translated **96** units to the right. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, where h , k , and a are constants, what is the value of $4a$?

ID: 1b2b20b9 Answer

Correct Answer:
28

Rationale

The correct answer is **28**. The equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - s)^2 = r^2$, where the center of the circle is (t, s) and the radius of the circle is r . It's given that circle A is defined by the equation $x^2 + (y - 6)^2 = 7$, which can be written as $(x - 0)^2 + (y - 6)^2 = (\sqrt{7})^2$. It follows that $r = \sqrt{7}$ and the radius of circle A is $\sqrt{7}$. It's also given that circle B has the same radius as circle A. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, then $a = r^2$. Substituting $\sqrt{7}$ for r in this equation yields $a = (\sqrt{7})^2$, or $a = 7$. It follows that the value of $4a$ is $4(7)$, or **28**.

Question Difficulty:
Hard

Question ID 2855cb58

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 2855cb58

A circle in the xy -plane has its center at $(16, 17)$ and has a radius of $7k$. Which equation represents this circle?

- A. $(x - 16)^2 + (y - 17)^2 = 49k$
- B. $(x - 16)^2 + (y - 17)^2 = 49k^2$
- C. $(x - 16)^2 + (y - 17)^2 = 7k$
- D. $(x - 16)^2 + (y - 17)^2 = 7k^2$

ID: 2855cb58 Answer

Correct Answer:

B

Rationale

Choice B is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that this circle has a center at $(16, 17)$ and a radius of $7k$. Substituting 16 for h , 17 for k , and $7k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 16)^2 + (y - 17)^2 = (7k)^2$, or $(x - 16)^2 + (y - 17)^2 = 49k^2$. Therefore, the equation that represents this circle is $(x - 16)^2 + (y - 17)^2 = 49k^2$.

Choice A is incorrect. This equation represents a circle with radius $7\sqrt{k}$, not $7k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Question Difficulty:

Hard

Question ID 9e44284b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 9e44284b

In the xy -plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

- A. 5
- B. 6.5
- C. $\sqrt{40}$
- D. $\sqrt{50}$

ID: 9e44284b Answer

Correct Answer:
A

Rationale

Choice A is correct. One way to find the radius of the circle is to rewrite the given equation in standard form, $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and the radius of the circle is r . To do this, divide the original equation, $2x^2 - 6x + 2y^2 + 2y = 45$, by 2 to make the leading coefficients of x^2 and y^2 each equal to 1: $x^2 - 3x + y^2 + y = 22.5$. Then complete the square to put the equation in standard form. To do so, first rewrite $x^2 - 3x + y^2 + y = 22.5$ as $(x^2 - 3x + 2.25) - 2.25 + (y^2 + y + 0.25) - 0.25 = 22.5$. Second, add 2.25 and 0.25 to both sides of the equation: $(x^2 - 3x + 2.25) + (y^2 + y + 0.25) = 25$. Since $x^2 - 3x + 2.25 = (x - 1.5)^2$, $y^2 + y + 0.25 = (y + 0.5)^2$, and $25 = 5^2$, it follows that $(x - 1.5)^2 + (y + 0.5)^2 = 5^2$. Therefore, the radius of the circle is 5.

Choices B, C, and D are incorrect and may be the result of errors in manipulating the equation or of a misconception about the standard form of the equation of a circle in the xy -plane.

Question Difficulty:
Hard

Question ID 24cec8d1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 24cec8d1

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 88° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

ID: 24cec8d1 Answer

Correct Answer:

46

Rationale

The correct answer is **46**. It's given that O is the center of a circle and that points R and S lie on the circle. Therefore, $\overline{OR}$ and $\overline{OS}$ are radii of the circle. It follows that $OR = OS$. If two sides of a triangle are congruent, then the angles opposite them are congruent. It follows that the angles $\angle RSO$ and $\angle ORS$, which are across from the sides of equal length, are congruent. Let x° represent the measure of $\angle RSO$. It follows that the measure of $\angle ORS$ is also x° . It's given that the measure of $\angle ROS$ is 88° . Because the sum of the measures of the interior angles of a triangle is 180° , the equation $x^\circ + x^\circ + 88^\circ = 180^\circ$, or $2x + 88 = 180$, can be used to find the measure of $\angle RSO$. Subtracting **88** from both sides of this equation yields $2x = 92$. Dividing both sides of this equation by **2** yields $x = 46$. Therefore, the measure of $\angle RSO$, in degrees, is **46**.

Question Difficulty:

Hard

Question ID fa2771d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: fa2771d5

Circle A has equation $(x - 7)^2 + (y + 3)^2 = 1$. In the xy -plane, circle B is obtained by translating circle A to the right **4** units. Which equation represents circle B?

- A. $(x - 7)^2 + (y + 7)^2 = 1$
- B. $(x - 3)^2 + (y + 3)^2 = 1$
- C. $(x - 11)^2 + (y + 3)^2 = 1$
- D. $(x - 7)^2 + (y - 1)^2 = 1$

ID: fa2771d5 Answer

Correct Answer:
C

Rationale

Choice C is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r units. It's given that circle A has the equation $(x - 7)^2 + (y + 3)^2 = 1$, which can be written as $(x - 7)^2 + (y - (-3))^2 = 1^2$. It follows that $h = 7$, $k = -3$, and $r = 1$. Therefore, the center of circle A is $(7, -3)$ and its radius is **1** unit. If circle A is translated **4** units to the right, the x -coordinate of the center will increase by **4**, while the y -coordinate and the radius of the circle will remain unchanged. Translating the center of circle A to the right **4** units yields $(7 + 4, -3)$, or $(11, -3)$. Therefore, the center of circle B is $(11, -3)$. Substituting **11** for h , -3 for k , and **1** for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 11)^2 + (y - (-3))^2 = 1$, or $(x - 11)^2 + (y + 3)^2 = 1$. Therefore, the equation $(x - 11)^2 + (y + 3)^2 = 1$ represents circle B.

Choice A is incorrect. This equation represents a circle obtained by shifting circle A down, rather than right, **4** units.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A left, rather than right, **4** units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A up, rather than right, **4** units.

Question Difficulty:
Hard

Question ID 3e577e4a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 3e577e4a

A circle in the xy -plane has its center at $(-4, -6)$. Line k is tangent to this circle at the point $(-7, -7)$. What is the slope of line k ?

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

ID: 3e577e4a Answer

Correct Answer:
A

Rationale

Choice A is correct. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It's given that the circle has its center at $(-4, -6)$ and line k is tangent to the circle at the point $(-7, -7)$. The slope of a radius defined by the points (q, r) and (s, t) can be calculated as $\frac{t-r}{s-q}$. The points $(-7, -7)$ and $(-4, -6)$ define the radius of the circle at the point of tangency. Therefore, the slope of this radius can be calculated as $\frac{(-6)-(-7)}{(-4)-(-7)}$, or $\frac{1}{3}$. If a line and a radius are perpendicular, the slope of the line must be the negative reciprocal of the slope of the radius. The negative reciprocal of $\frac{1}{3}$ is -3 . Thus, the slope of line k is -3 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the slope of the radius of the circle at the point of tangency, not the slope of line k .

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID ab176ad6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: ab176ad6

The equation $(x + 6)^2 + (y + 3)^2 = 121$ defines a circle in the xy -plane. What is the radius of the circle?

ID: ab176ad6 Answer

Rationale

The correct answer is 11. A circle with equation $(x - a)^2 + (y - b)^2 = r^2$, where a , b , and r are constants, has center (a, b) and radius r . Therefore, the radius of the given circle is $\sqrt{121}$, or 11.

Question Difficulty:
Hard

Question ID 249d3f80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 249d3f80

Point O is the center of a circle. The measure of arc RS on this circle is 100° . What is the measure, in degrees, of its associated angle ROS ?

ID: 249d3f80 Answer

Correct Answer:
100

Rationale

The correct answer is **100**. It's given that point O is the center of a circle and the measure of arc RS on the circle is 100° . It follows that points R and S lie on the circle. Therefore, $\overline{OR}$ and $\overline{OS}$ are radii of the circle. A central angle is an angle formed by two radii of a circle, with its vertex at the center of the circle. Therefore, $\angle ROS$ is a central angle. Because the degree measure of an arc is equal to the measure of its associated central angle, it follows that the measure, in degrees, of $\angle ROS$ is **100**.

Question Difficulty:
Hard

Question ID b0a72bdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: b0a72bdc

What is the diameter of the circle in the xy -plane with equation $(x - 5)^2 + (y - 3)^2 = 16$?

- A. 4
- B. 8
- C. 16
- D. 32

ID: b0a72bdc Answer

Correct Answer:

B

Rationale

Choice B is correct. The standard form of an equation of a circle in the xy -plane is $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the length of the radius of the circle is r . For the circle in the xy -plane with equation $(x - 5)^2 + (y - 3)^2 = 16$, it follows that $r^2 = 16$. Taking the square root of both sides of this equation yields $r = 4$ or $r = -4$. Because r represents the length of the radius of the circle and this length must be positive, $r = 4$. Therefore, the radius of the circle is 4. The diameter of a circle is twice the length of the radius of the circle. Thus, $2(4)$ yields 8. Therefore, the diameter of the circle is 8.

Choice A is incorrect. This is the radius of the circle.

Choice C is incorrect. This is the square of the radius of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID b8a225ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: b8a225ff

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

ID: b8a225ff Answer

Correct Answer:
16

Rationale

The correct answer is **16**. An equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u) , the radius of the circle is r , and where t, u , and r are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 4$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 2^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is **2**. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is **2(2)**, or **4**. Substituting -5 for t , 5 for u , and **4** for r into the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 4^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 16$. It follows that the equation of circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = 16$. Therefore, the value of k is **16**.

Question Difficulty:
Hard

Question ID ebbf23ae

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: ebbf23ae

A circle in the xy -plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r ?

ID: ebbf23ae Answer

Correct Answer:
5

Rationale

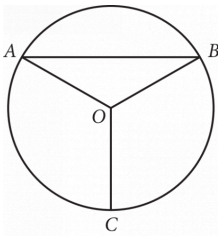
The correct answer is **5**. The standard form of an equation of a circle in the xy -plane is $(x - h)^2 + (y - k)^2 = r^2$, where h , k , and r are constants, the coordinates of the center of the circle are (h, k) , and the length of the radius of the circle is r . It's given that an equation of the circle is $(x - 2)^2 + (y - 9)^2 = r^2$. Therefore, the center of this circle is $(2, 9)$. It's given that the endpoints of a diameter of the circle are $(2, 4)$ and $(2, 14)$. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. Substituting the center of the circle $(2, 9)$ and one endpoint of the diameter $(2, 4)$ in this formula gives a distance of $\sqrt{(2 - 2)^2 + (9 - 4)^2}$, or $\sqrt{0^2 + 5^2}$, which is equivalent to **5**. Since the distance from the center of the circle to an endpoint of a diameter is **5**, the value of r is **5**.

Question Difficulty:
Hard

Question ID 69b0d79d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 69b0d79d



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

- A. 9π
- B. 12π
- C. 15π
- D. 18π

ID: 69b0d79d Answer

Correct Answer:
B

Rationale

Choice B is correct. Because segments OA and OB are radii of the circle centered at point O , these segments have equal lengths. Therefore, triangle AOB is an isosceles triangle, where angles OAB and OBA are congruent base angles of the triangle. It's given that angle OAB measures 30° . Therefore, angle OBA also measures 30° . Let x° represent the measure of angle AOB . Since the sum of the measures of the three angles of any triangle is 180° , it follows that $30^\circ + 30^\circ + x^\circ = 180^\circ$, or $60^\circ + x^\circ = 180^\circ$.

Subtracting 60° from both sides of this equation yields $x^\circ = 120^\circ$, or $\frac{2\pi}{3}$ radians. Therefore, the measure of angle AOB , and thus the measure of arc $\widehat{AB}$, is $\frac{2\pi}{3}$ radians. Since $\overline{OC}$ is a radius of the given circle and its length is 18, the length of the radius of the circle is 18. Therefore, the length of arc $\widehat{AB}$ can be calculated as $\left(\frac{2\pi}{3}\right)(18)$, or 12π .

Choices A, C, and D are incorrect and may result from conceptual or computational errors.

Question Difficulty:

Question ID 2266984b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 2266984b

$x^2 + 20x + y^2 + 16y = -20$

The equation above defines a circle in the xy -plane. What are the coordinates of the center of the circle?

- A. $(-20, -16)$
- B. $(-10, -8)$
- C. $(10, 8)$
- D. $(20, 16)$

ID: 2266984b Answer

Correct Answer:
B

Rationale

Choice B is correct. The standard equation of a circle in the xy -plane is of the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) are the coordinates of the center of the circle and r is the radius. The given equation can be rewritten in standard form by completing the squares. So the sum of the first two terms, $x^2 + 20x$, needs a 100 to complete the square, and the sum of the second two terms, $y^2 + 16y$, needs a 64 to complete the square. Adding 100 and 64 to both sides of the given equation yields $(x^2 + 20x + 100) + (y^2 + 16y + 64) = -20 + 100 + 64$, which is equivalent to $(x + 10)^2 + (y + 8)^2 = 144$. Therefore, the coordinates of the center of the circle are $(-10, -8)$.

Choices A, C, and D are incorrect and may result from computational errors made when attempting to complete the squares or when identifying the coordinates of the center.

Question Difficulty:
Hard

Question ID e50afdd3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: e50afdd3

$$(x + 4)^2 + (y - 19)^2 = 121$$

The graph of the given equation is a circle in the xy -plane. The point (a, b) lies on the circle. Which of the following is a possible value for a ?

- A. -16
- B. -14
- C. 11
- D. 19

ID: e50afdd3 Answer

Correct Answer:

B

Rationale

Choice B is correct. An equation of the form $(x - h)^2 + (y - k)^2 = r^2$, where h , k , and r are constants, represents a circle in the xy -plane with center (h, k) and radius r . Therefore, the circle represented by the given equation has center $(-4, 19)$ and radius 11 . Since the center of the circle has an x -coordinate of -4 and the radius of the circle is 11 , the least possible x -coordinate for any point on the circle is $-4 - 11$, or -15 . Similarly, the greatest possible x -coordinate for any point on the circle is $-4 + 11$, or 7 . Therefore, if the point (a, b) lies on the circle, it must be true that $-15 \leq a \leq 7$. Of the given choices, only -14 satisfies this inequality.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID 35d37640

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 35d37640

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

ID: 35d37640 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the circle is a unit circle, which means the circle has a radius of 1 . It's also given that point G is the center of the circle and has coordinates $(0, 0)$ and that point H lies on the circle and has coordinates $(-1, y)$. Since the radius of the circle is 1 , the value of y must be 0 , as all other points with an x -coordinate of -1 are a distance greater than 1 away from point G . Since F and H are points on the unit circle centered at G , let side FG be the starting side of the angle and side GH be the terminal side of the angle. Since side FG is on the positive x -axis and side GH is on the negative x -axis, side FG is half of a rotation around the unit circle, or π radians, away from side GH . Therefore, the positive measure of angle FGH , in radians, is equal to π plus an integer multiple of 2π . In other words, the positive measure of angle FGH , in radians, is an odd integer multiple of π . Of the given choices, only 25π is an odd integer multiple of π .

Choice A is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0, -1)$, not $(-1, 0)$.

Choice B is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0, 1)$, not $(-1, 0)$.

Choice C is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(1, 0)$, not $(-1, 0)$.

Question Difficulty:
Hard

Question ID 76c73dbf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 76c73dbf

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. What is the length of the circle's radius?

ID: 76c73dbf Answer

Correct Answer:
10

Rationale

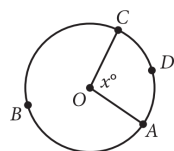
The correct answer is **10**. It's given that the graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the length of the radius of the circle is r . The term $(x - h)^2$ in this equation can be obtained by adding the square of half the coefficient of x to both sides of the given equation to complete the square. The coefficient of x is **1**. Half the coefficient of x is $\frac{1}{2}$. The square of half the coefficient of x is $\frac{1}{4}$. Adding $\frac{1}{4}$ to each side of $(x^2 + x) + (y^2 + y) = \frac{199}{2}$ yields $(x^2 + x + \frac{1}{4}) + (y^2 + y) = \frac{199}{2} + \frac{1}{4}$, or $(x + \frac{1}{2})^2 + (y^2 + y) = \frac{199}{2} + \frac{1}{4}$. Similarly, the term $(y - k)^2$ can be obtained by adding the square of half the coefficient of y to both sides of this equation, which yields $(x + \frac{1}{2})^2 + (y^2 + y + \frac{1}{4}) = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$, or $(x + \frac{1}{2})^2 + (y + \frac{1}{2})^2 = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$. This equation is equivalent to $(x + \frac{1}{2})^2 + (y + \frac{1}{2})^2 = 100$, or $(x + \frac{1}{2})^2 + (y + \frac{1}{2})^2 = 10^2$. Therefore, the length of the circle's radius is **10**.

Question Difficulty:
Hard

Question ID c8345903

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: c8345903



The circle above has center O , the length of arc $\widehat{ADC}$ is 5π , and

$x = 100$. What is the length of arc $\widehat{ABC}$?

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

ID: c8345903 Answer

Correct Answer:
B

Rationale

Choice B is correct. The ratio of the lengths of two arcs of a circle is equal to the ratio of the measures of the central angles that subtend the arcs. It's given that arc $\widehat{ADC}$ is subtended by a central angle with measure 100° . Since the sum of the measures of the angles about a point is 360° , it follows that arc $\widehat{ABC}$ is subtended by a central angle with measure $360^\circ - 100^\circ = 260^\circ$. If s is the length of arc $\widehat{ABC}$, then s must satisfy the ratio $\frac{s}{5\pi} = \frac{260}{100}$. Reducing the fraction $\frac{260}{100}$ to its simplest form gives $\frac{13}{5}$. Therefore, $\frac{s}{5\pi} = \frac{13}{5}$. Multiplying both sides of $\frac{s}{5\pi} = \frac{13}{5}$ by 5π yields $s = 13\pi$.

Choice A is incorrect. This is the length of an arc consisting of exactly half of the circle, but arc $\widehat{ABC}$ is greater than half of the circle. Choice C is incorrect. This is the total circumference of the circle. Choice D is incorrect. This is half the length of arc $\widehat{ABC}$, not its full length.

Question Difficulty:
Hard

Question ID 9adb86ed

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 9adb86ed

Points Q and R lie on a circle with center P . The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\overline{QR}$?

- A. $13\sqrt{2}$
- B. 13
- C. $9\sqrt{2}$
- D. 9

ID: 9adb86ed Answer

Correct Answer:
B

Rationale

Choice B is correct. Since it's given that P is the center of a circle with a radius of 9 inches, and that points Q and R lie on that circle, it follows that $\overline{PQ}$ and $\overline{RP}$ of triangle PQR each have a length of 9 inches. Let the length of $\overline{QR}$ be x inches. It follows that the perimeter of triangle PQR is $9 + 9 + x$ inches. Since it's given that the perimeter of triangle PQR is 31 inches, it follows that $9 + 9 + x = 31$, or $18 + x = 31$. Subtracting 18 from both sides of this equation gives $x = 13$. Therefore, the length, in inches, of $\overline{QR}$ is 13 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 858fd1cf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 858fd1cf

A circle in the xy -plane has its center at $(-1, 1)$. Line t is tangent to this circle at the point $(5, -4)$. Which of the following points also lies on line t ?

- A. $(0, \frac{6}{5})$
- B. $(4, 7)$
- C. $(10, 2)$
- D. $(11, 1)$

ID: 858fd1cf Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the circle has its center at $(-1, 1)$ and that line t is tangent to this circle at the point $(5, -4)$. Therefore, the points $(-1, 1)$ and $(5, -4)$ are the endpoints of the radius of the circle at the point of tangency. The slope of a line or line segment that contains the points (a, b) and (c, d) can be calculated as $\frac{d-b}{c-a}$. Substituting $(-1, 1)$ for (a, b) and $(5, -4)$ for (c, d) in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-1}{5-(-1)}$, or $-\frac{5}{6}$. Thus, the slope of this radius is $-\frac{5}{6}$. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It follows that line t is perpendicular to the radius at the point $(5, -4)$, so the slope of line t is the negative reciprocal of the slope of this radius. The negative reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line t is $\frac{6}{5}$. Since the slope of line t is the same between any two points on line t , a point lies on line t if the slope of the line segment connecting the point and $(5, -4)$ is $\frac{6}{5}$. Substituting choice C, $(10, 2)$, for (a, b) and $(5, -4)$ for (c, d) in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-2}{5-10}$, or $\frac{6}{5}$. Therefore, the point $(10, 2)$ lies on line t .

Choice A is incorrect. The slope of the line segment connecting $(0, \frac{6}{5})$ and $(5, -4)$ is $\frac{-4-\frac{6}{5}}{5-0}$, or $-\frac{26}{25}$, not $\frac{6}{5}$.

Choice B is incorrect. The slope of the line segment connecting $(4, 7)$ and $(5, -4)$ is $\frac{-4-7}{5-4}$, or -11 , not $\frac{6}{5}$.

Choice D is incorrect. The slope of the line segment connecting $(11, 1)$ and $(5, -4)$ is $\frac{-4-1}{5-11}$, or $\frac{5}{6}$, not $\frac{6}{5}$.

Question Difficulty:
Hard